FLEET TEAM DIGEST

737MAX-FTD-34-17005

Issue Title : UNEXPECTED REMOVAL OF FLIGHT DIRECTOR (F/D), BIASED OUT OF VIEW

Airplane Model	ATA	Minor Model(s)
737MAX	3433-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
12/12/2017	12/11/2020	11/30/2020

Estimated Completion Date	Next Update Date

Revision Description

Updated Status, Final Action, Milestones, and Operator Action sections and closed FTD.

Applicability

All 737NG/MAX airplane models

Description

Boeing has received reports from some operators of unexpected removal of flight director, biased out of view. This can also be accompanied by Autopilot disconnect, and/or NO AUTOLAND/NO LAND 3.

Background

Some operators have experienced a loss of Flight Director, Biased Out of View (FD/BOV) and Autopilot disconnects on approach.

Aircraft certified for fail-operational autoland (Cat IIIb) may see NO AUTOLAND or NO LAND 3 annunciations.

Operators have also experienced the ‘STALL WARNING SYS L’ and/or ‘STALL WARNING SYS R’ latched status messages displayed on the OMF (reference Fleet Team Digest 737MAX-FTD-27-18002).

In addition, if the flight crew ARM LNAV on the ground and the LNAV FMA flashes on the PFD/MDS, this can indicate that the FCC’s are not receiving radio altimeter data from one of the RA’s

Status

1. This issue is seen most often in the interface between the Right-side LRRA and FCC-B, but can also be seen between either side LRRA and one or both FCC’s. It has also been seen to impact data going to the Spoiler Control Electronics (SCE).

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2. There are two LRRA data busses:

- Data bus 1 supplies data to the Flight Control Computer (FCC) and Spoiler Control Electronics (SCE).

- Data bus 2 supplies data to the DEU, Digital Flight Data Acquisition Unit (DFDAU), Ground Proximity Warning System (GPWS), Traffic Collision Avoidance System (TCAS).

FDR data does not show LRRA data drops even when FD/BOV, Autopilot disconnects, or No Autoland indications are observed.

This can occur due to an internal health monitor in the LRRA that will disable the output to the FCC's and SCE on Databus 1 when a fault is detected in the unit for a certain time period. Under this condition, the altitude output to the FCC's and SCE is disabled until the failed LRRA power is cycled (the LRRA will continue to output altitude data on Databus 2 to the FDR, DEU (Display Electronics Unit), DFDAU, GPWS, and TCAS).

3. Recent investigation and troubleshooting indicates that this condition can be caused by an intermittent LRRA coax cable attachment, including the following:

-the clamp that holds the coax cable in place where it exits the back of the Avionics Rack pulling on the cable and causing intermittent contact with the LRRA rack connector

-the coax cable / antenna connector being loose, mis-aligned, mis-pinned, or damaged, causing intermittent contact with the antennas

-this condition can also be caused by a faulty/damaged coax cable, or a faulty transmit or receive antenna, either of which may require replacement of the part

See the '737MAX FTD Attachment' for recommended troubleshooting steps.

4. Boeing Production and Quality is conducting investigations into manufacturing, installation, wire routing, and design in order to determine the root cause(s) of these issues. These investigations are focusing on the following to determine/implement corrective action for this issue:

–Modification of coax form board and winding tool to ensure bundles and connectors maintain correct orientation

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–Audit of coax supplier resulted in manufacturing process updates and additional tests to ensure cable build quality

–Implemented connector covers at coax cable / antenna interface to prevent post installation damage

–Implemented cable holding clamp change to prevent tension on coax cable connectors behind the LRU rack

Interim Action

If BITE test of the LRRA system indicates no fault, and/or:

-The FCC indicates any combination of maintenance messages 22-11051 / 22-11053 (no data from RA1) and 22-11052 / 22-11054 (no data from RA2), 22-11385 (invalid data from RA1) / 22-11425 (invalid data from RA2), 22-11270 / 22-11272 (related to NO LAND 3 faults), and/or,

-The SCE indicates any combination of maintenance messages 27-02040 (no data from RA1) / 27-02160 (invalid data from RA1) and 27-02050 (no data from RA2) / 27-02170 (invalid data from RA2), and/or,

-The SCE/DPC indicates any combination of maintenance messages 34-90021 (no data from RA1) and 34-90022 (no data from RA2), and/or,

-The SMYD indicates any combination of maintenance messages 27-31021 (SMYD1 detects invalid DPC data, possibly due to RA1) and 27-32021 (SMYD2 detects invalid DPC data, possibly due to RA1),

then there could be a potential issue with one or more of the following:

1) Coax cable holding clamp issue resulting in the coax cable at the back of the LRRA Transceiver mounting tray being pulled too tight by the clamp, and causing intermittent connection with the transceiver due to airplane vibration and flexing in flight

2) Connector interface issue between the coax cable and the transmit or receive antennas (loose, misaligned, faulty) causing intermittent connection with the antenna(s) due to airplane vibration and flexing in flight

3) Issue with the coaxial cable itself running between the LRRA transceiver tray and the transmit and/or receive antenna(s)

4) Issue with the transmit and/or receive antenna(s) itself

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Follow the troubleshooting guidance in the '737MAX FTD Attachment' in order to better characterize the fault condition.

FDR data can also be used to potentially identify which side LRRA is impacted and the specific antenna(s) that have the intermittent connection issue.

Final Action

Refer to FIM troubleshooting instructions associated with the FCC indicating any combination of maintenance messages 22-11051 / 22-11053 (no data from RA1) and 22-11052 / 22-11054 (no data from RA2), 22-11385 (invalid data from RA1) / 22-11425 (invalid data from RA2), 22-11270 / 22-11272 (related to NO LAND 3 faults), and/or,

-The SCE indicates any combination of maintenance messages 27-02040 (no data from RA1) / 27-02160 (invalid data from RA1) and 27-02050 (no data from RA2) / 27-02170 (invalid data from RA2), and/or,

-The SCE/DPC indicates any combination of maintenance messages 34-90021 (no data from RA1) and 34-90022 (no data from RA2), and/or,

-The SMYD indicates any combination of maintenance messages 27-31021 (SMYD1 detects invalid DPC data, possibly due to RA1) and 27-32021 (SMYD2 detects invalid DPC data, possibly due to RA1).

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Flight Control Computer FIM troubleshooting instructions updated - October 2019

Operator Action

See 737MAX FTD Attachment and Interim Action for recommended actions.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-27-18002		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- In Service Occurrences : null

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- Manufacturing Quality Issues : Issues related to airplane or component manufacturing quality

Attachments:

a) FTD Attachment -

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